From Noise to Narrative

A Visualization Tool for DDoS Events Across PEST Dimensions

Jesse van Dijk
Business Information Technology
s2920174
j.vandijk-9@student.utwente.nl

Introduction

- Current affairs: Recent attacks on Israel, European Parliament (2023)
- Situation: DDoS attacks are relevant threats, and are actively being reported on
- Complicatie: it is hard to gain insight into the impact of these dimensions
- Question: How do we measure and visualize different dimensions of impact of a DDoS event by clustering article-level news data?

Research Questions

- RQ1: To what extent does news coverage of a DDoS event reflect its PEST-dimension impact?
- RQ2: How effectively can event-level groupings of DDoS events be constructed from article-level news data using semantic embedding and clustering?
- RQ3: How do we create a measurement tool to visualize the impact of an event based on the annotated dataset?

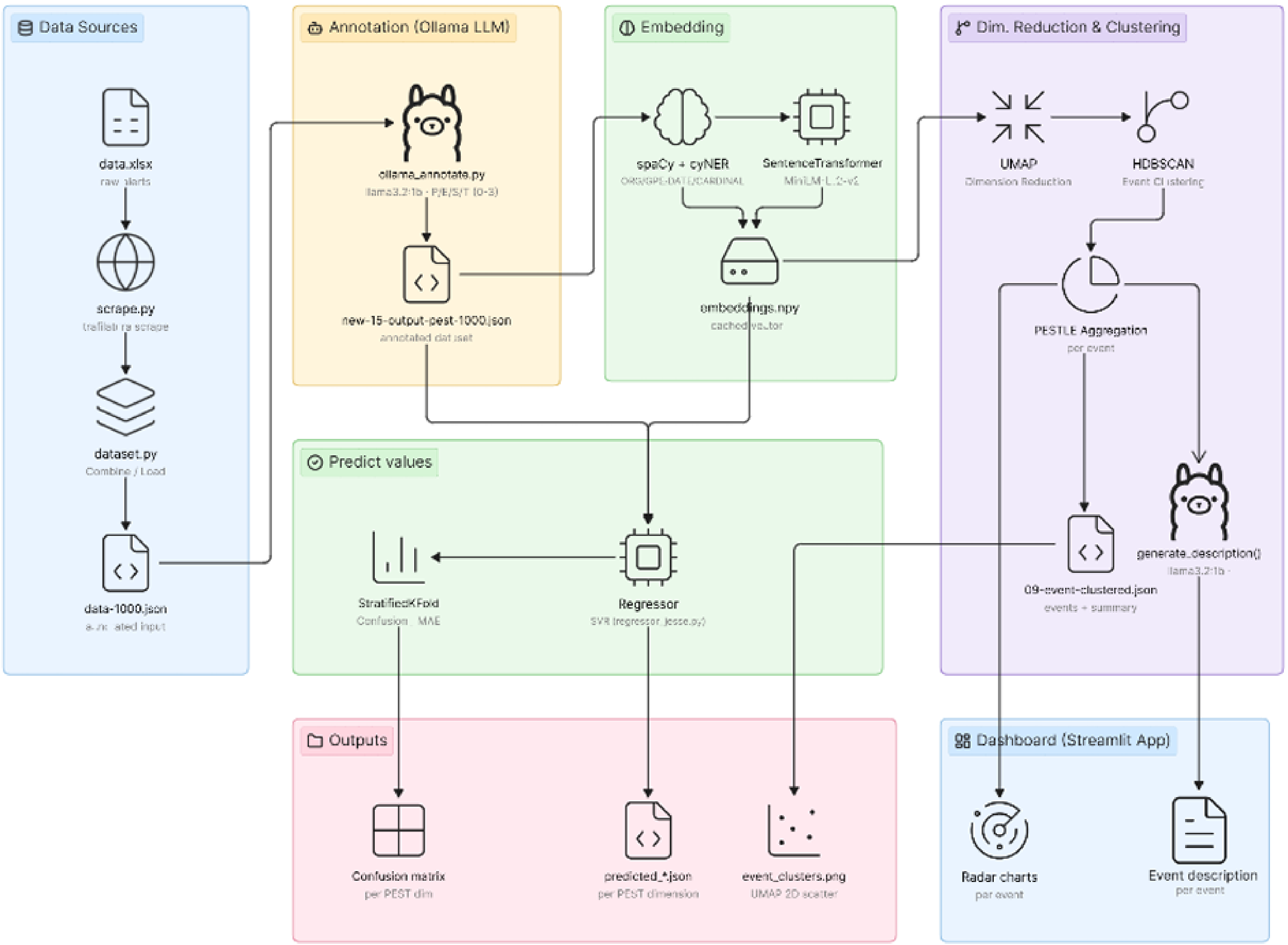
PEST

- Political
- Economical
- Social
- Technological

Table 2: Operationalization of PEST dimensions for cyber incident impact assessment

Dimension	Literature-based definition	Observed cyber impact characteristics
Political	National stability, geopolitical tension [8][5]. Some states even employ cyber attacks as part of broader political and strategic objectives [7].	Cyber attacks as tools in geopolitical rivalry and hybrid warfare; influence on national security agendas, international norms, and political response mechanisms.
Economical	Reputational damage, revenue loss, recovery costs, regulatory fines, and declining stock prices [2].	Macro-level modelling of extreme cyber-event losses; firm-level and sectoral financial damage; disruptions to critical infrastructure such as ports and energy systems.
Social	Societal cohesion and public distress [8].	Psychological distress, erosion of trust, weakened organisational cohesion, and societal disruption caused by attacks on critical infrastructure.
Technological	Physical and digital harm, including infrastructure disruption [2].	Impacts on availability, integrity, and confidentiality; operational inoperability of critical infrastructure; increasing socio-technical sophistication of attacks.

Architecture



Data + annotation

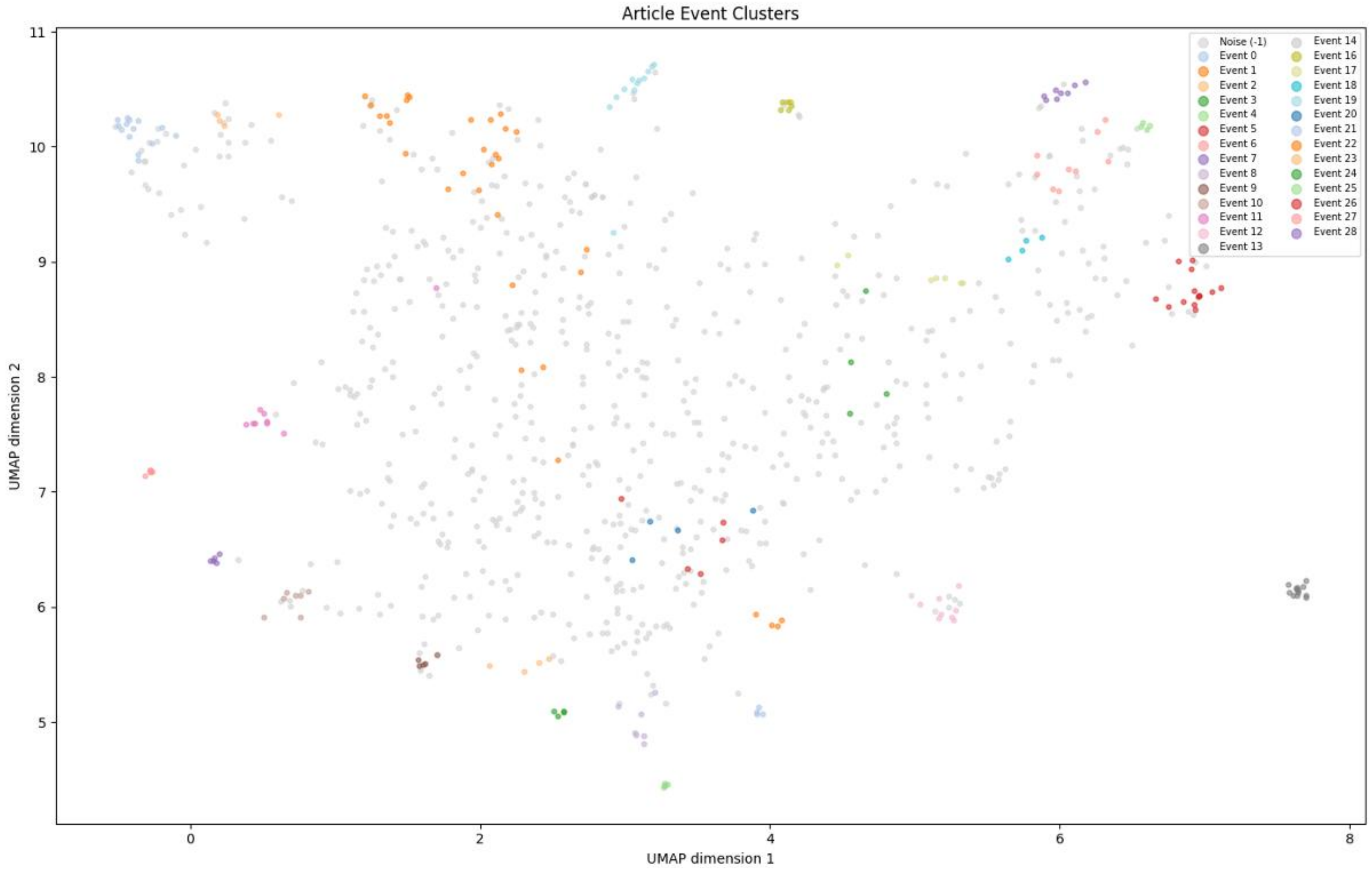
- Complete database of 235913 articles
- Between 20-8-2015 and 10-5-2023
- 1000 selected, 860 DDoS relevant

Table 3: Distribution of annotation scores across the four dimensions.

Dimension	Score 0	Score 1	Score 2	Score 3
Political	569 (66.2%)	16 (1.9%)	223 (25.9%)	52 (6.0%)
Economical	549 (63.8%)	79 (9.2%)	162 (18.8%)	70 (8.1%)
Social	637 (74.1%)	136 (15.8%)	61 (7.1%)	26 (3.0%)
Technological	184 (21.4%)	49 (5.7%)	162 (18.8%)	465 (54.1%)

Clustering

- UMAP
- HDBSCAN
  - Min_cluster_size
  - Min_samples
  - Max_cluster_size



Regressor

- Goal: Test if PEST impact is learnable
- SVR with nested cross-validation
- Label noise sets upper performance bound

Table 5: Cross-validation performance metrics across training dimensions.

Dimension	MAE	RMSE	Spearman
Political	0.852 ± 0.057	1.159	0.188
Economical	0.874 ± 0.045	1.152	0.140
Social	0.592 ± 0.029	0.928	-0.003
Technological	1.023 ± 0.036	1.302	0.140

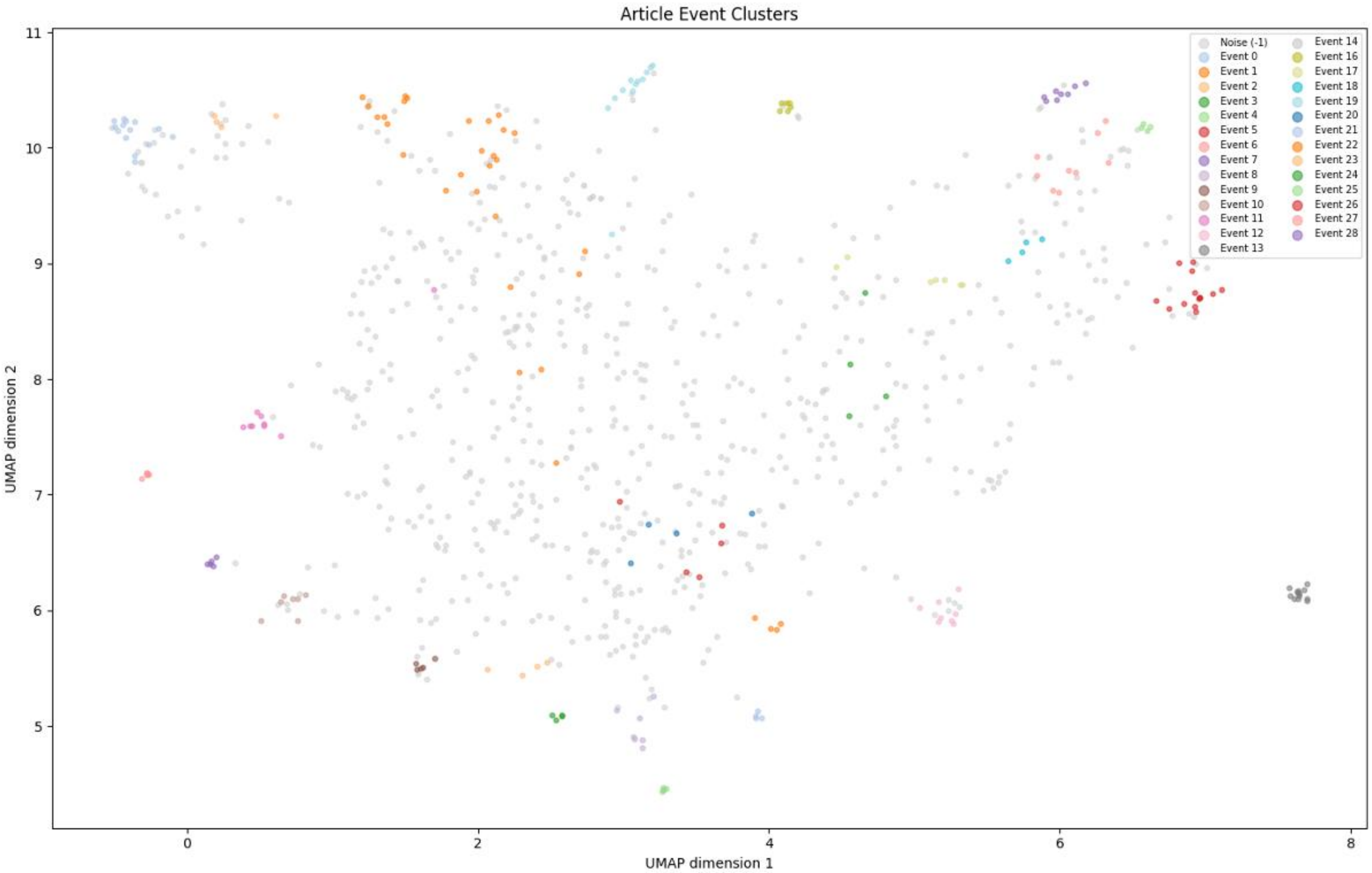
Results 1

Table 4: Performance metrics and bias direction acros ing dimensions.

Dimension	MAE	Cohen's κ	Bias Direction
Political	1.03	0.164	-0.39
Economical	0.96	0.207	+0.29
Social	0.85	0.141	-0.59
Technological	1.43	0.057	+1.03

		Pred 0	Pred 1	Pred 2	Pred 3
Act 0	P	233	276	59	1
	E	193	307	45	4
	S	356	275	6	0
	T	3	44	109	28
Act 1	P	8	8	0	0
	E	29	43	6	1
	S	69	63	1	3
	T	2	6	35	6
Act 2	P	59	109	45	10
	E	32	101	26	3
	S	32	29	0	0
	T	1	18	108	35
Act 3	P	4	30	18	0
	E	7	52	9	2
	S	10	16	0	0
	T	10	59	282	114

Results 2



28 events,
860 articles,
75.8% noise

Conclusion 1

- LLM annotation is not a reliable substitute for human judgment ($\kappa = 0.057$–0.207)
- SVR cannot predict PEST scores from embeddings alone (Spearman ≈ 0–0.19)
- Label noise from the LLM likely explains poor regression performance
- $K < 0.2$ = poor

Conclusion 2

- ‘Correctly’ identify 28 distinct events from 860 articles
  - Manual validation through each event, subset discussed in paper
- Entity-enriched embeddings + HDBSCAN successfully clusters related articles
- 75.8% noise is high, but is a deliberate tradeoff favoring precision over recall

Conclusion 3

- Artifact demonstrates automated event-level impact analysis is feasible without manual labelling
- Pipeline combines LLM annotation → semantic embedding → clustering → radar chart visualization
- Reliability bounded by quality of LLM annotations
- Published openly on GitHub

Artifact showcase

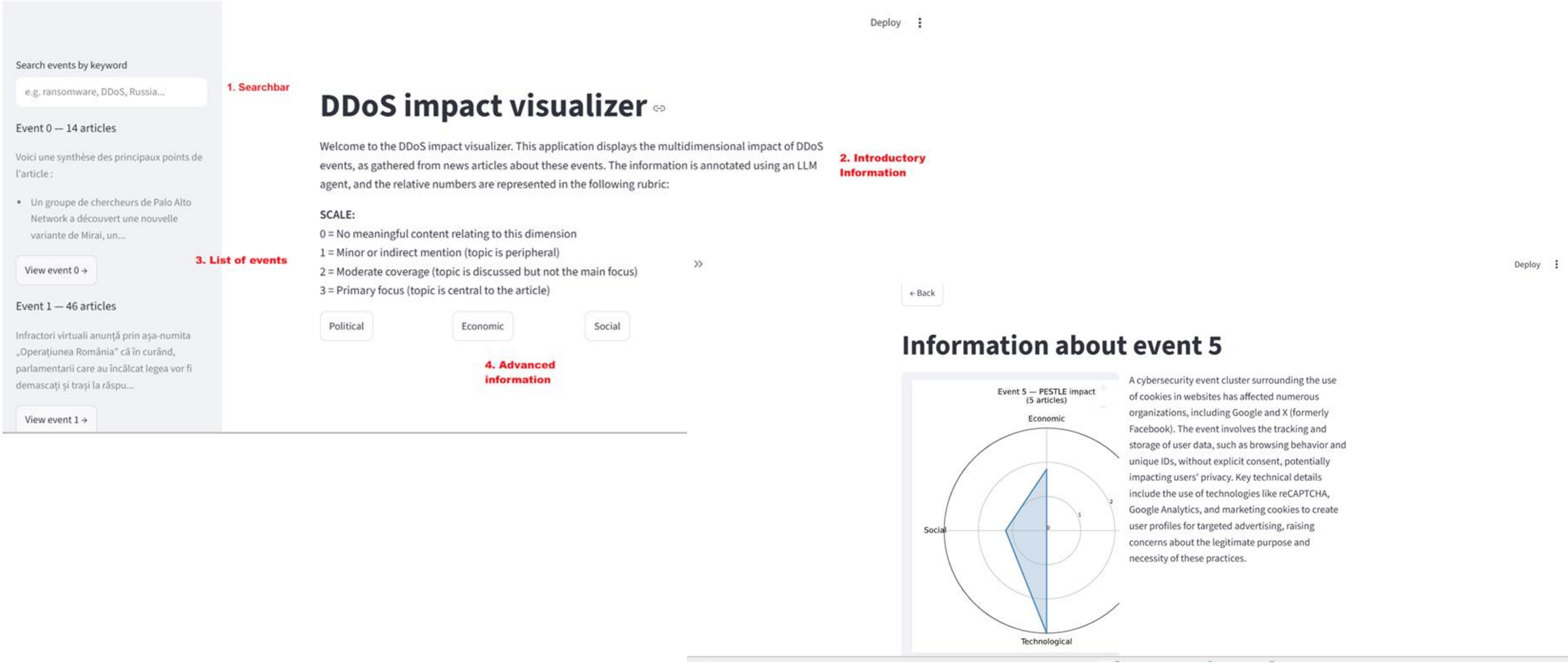


Figure 5: Event overview

Limitations + future work

- Regression model comparison
- Llama 3.2 quality vs. larger models
- Increase entity enrichment
- Single out attack news instead of all DDoS events